DK-DDIL: Adaptive Knowledge Retention for Dynamic Domain-Incremental Learning in Medical Imaging

Yuxi Ma[1*] Sujie Liu[2*] Jing Yang[1] Jiacheng Wang[3]
Yiping Chen[4†] Baptiste Magnier[5] Liansheng Wang[1,2†]

[1] National Institute for Data Science in Health and Medicine, Xiamen University, Xiamen, China
[2] Department of Computer Science at the School of Informatics, Xiamen University, Xiamen, China
[3] Manteia Technology, Co., Ltd., Xiamen, China
[4] School of Geospatial Engineering and Science, Sun Yat-Sen University, Guangzhou, China
[5] EuroMov Digital Health in Motion, Univ Montpellier, IMT Mines Ales, Ales, France

Abstract

Large-scale foundation models pretrained on massive datasets have demonstrated strong generalization capabilities in medical image analysis. However, they are typically trained on static datasets and struggle to cope with the continuously evolving nature of clinical data, where new imaging devices, institutions, and disease subtypes constantly emerge. While domain-incremental learning (DIL) provides a solution for sequential adaptation without revisiting historical data, existing methods typically assume fixed label spaces and limited domain heterogeneity, restricting their applicability to real-world clinical scenarios. To address these challenges, we propose ***DK-DDIL****, a rehearsal-free framework for dynamic DIL that integrates two synergistic modules: a* ***Dynamic Adaptation Module (DAM)*** *employing dynamic rank selection and adaptive regularization to flexibly allocate model capacity under domain shifts, and a* ***Knowledge Inheritance and Refinement (KIR)*** *module that stabilizes cross-domain knowledge transfer through selective adapter fusion and prototype-level contrastive refinement. Experiments on the Skin Pathology Diagnosis dataset, the Cyst-X 3D MRI cohort, and the OfficeHome benchmark demonstrate that DK-DDIL consistently outperforms state-of-the-art DIL approaches, highlighting its effectiveness and versatility across dynamic 2D medical, 3D medical, and natural image domains.*

1. Introduction

Foundation models have shown remarkable generalization in medical image analysis [36]. However, they are typically trained on static and closed datasets, which fail to capture the continuously evolving nature of real-world clinical data [1]. In practice, medical images are generated over time and across institutions, exhibiting variations in imaging devices, acquisition protocols, patient demographics, and disease subtypes. Such distributional shifts often cause severe degradation in model performance when deployed to unseen domains [7, 14, 16]. Moreover, retraining large-scale foundation models is computationally prohibitive [22], and privacy regulations further restrict inter-institutional data sharing [25, 34]. These constraints underscore the pressing need for models that can *continuously adapt to new domains without revisiting historical data*, aligning with realistic clinical deployment scenarios.

To address these dynamic and non-stationary settings, recent studies have explored *domain-incremental learning* (DIL). DIL is a learning paradigm that allows models to adapt to new domains or datasets sequentially without forgetting previously learned knowledge. Existing DIL approaches are mainly categorized into two families: **prototype-based** methods [18, 40, 41] and **prompt-based** methods [26, 32, 33]. Although these methods achieve strong performance in conventional DIL settings, they generally assume a fixed label space and pre-defined domain boundaries, which limits their applicability in realistic dynamic DIL scenarios. In dynamic DIL, not only do data distributions and task complexities evolve over time, but the label space itself may also expand as new classes are introduced. This dynamic nature creates significant challenges because models must maintain a careful stability–plasticity balance, adaptively allocate capacity across domains of varying complexity, and effectively consolidate knowledge between previously seen and newly introduced classes.

Motivated by this gap, we develop **DK-DDIL**, a

*These authors contributed equally to this work.
†Corresponding authors.

rehearsal-free *Dynamic DIL* framework built upon Low-Rank Adaptation (LoRA) [11]. While LoRA enables efficient fine-tuning through lightweight low-rank adapters, its conventional design remains suboptimal for dynamic DIL: fixed-rank adapters lack flexibility across domains of varying intrinsic complexity [31]; and independently trained adapters can interfere with shared representations, destabilizing cross-domain learning [35]. To address these issues, DK-DDIL introduces two synergistic modules that unify adaptive parameter tuning and cross-domain knowledge consolidation: the **Dynamic Adaptation Module (DAM)**, which adaptively regulates LoRA ranks via learnable masks, scaling factors, and sparsity regularization, enabling domain-aware capacity scaling; and the **Knowledge Inheritance and Refinement (KIR)** mechanism, which preserves cross-domain consistency through selective adapter fusion and prototype-level contrastive refinement. Together, these modules enable flexible adaptation to evolving domains while mitigating catastrophic forgetting in a rehearsal-free manner. We evaluate DK-DDIL on three representative benchmarks covering diverse domain dynamics: **Skin Pathology Diagnosis**, a temporally evolving dermoscopic dataset aggregated from multiple public sources; **Cyst-X**, a clinical multi-center 3D MRI cohort for IPMN risk stratification; and **OfficeHome**, a standard natural-image benchmark for DIL. These datasets collectively span temporal, multi-center, and distributional domain shifts, providing a comprehensive evaluation of DK-DDIL across medical and natural image scenarios.

Our main contributions are summarized as follows: (*i*) We introduce a rehearsal-free dynamic DIL framework that jointly handles **evolving label spaces** and **heterogeneous domain shifts**, a setting rarely explored in prior DIL studies, especially under privacy constraints typical of clinical data. (*ii*) We design a DAM that adaptively allocates LoRA capacity through learnable rank masks, and a KIR mechanism that preserves both parameter- and prototype-level consistency. This unified mechanism ensures domain-aware adaptation while mitigating catastrophic forgetting. (*iii*) We conduct extensive experiments across three complementary domains: the Skin Pathology Diagnosis dataset, the Cyst-X 3D MRI cohort, and the OfficeHome benchmark, demonstrating the effectiveness, generality, and efficiency of our method under realistic non-stationary settings.

2. Related Work

2.1. Pretrained Models (PTMs) for DIL

Large-scale pretrained models (PTMs), such as ViT and CLIP, have demonstrated strong generalization and have become the backbone for many DIL methods. However, directly fine-tuning PTMs often results in catastrophic forgetting and imposes substantial computational overhead. Consequently, recent PTM-based DIL research has evolved along two major lines: *prompt-based* and *adapter-based* approaches.

Prompt-based methods guide PTMs through learnable prompt tokens while keeping the backbone frozen or lightly tuned. L2P [33] pioneers the use of prompts for continual learning by maintaining a learnable prompt pool for task-specific retrieval, thereby mitigating catastrophic forgetting. DualPrompt [32] extends this idea with general and task-specific prompts to improve cross-domain transferability, while CODA-Prompt [26] further improves flexibility through dynamic prompt assembly via conditional attention. Although prompt-based methods achieve excellent parameter efficiency and are suitable for rehearsal-free scenarios, their performance remains constrained by limited representational capacity and fixed retrieval structures.

Adapter-based approaches insert lightweight trainable modules into frozen PTMs, enabling efficient adaptation with minimal additional parameters. EASE [38] constructs low-dimensional subspaces for each task and integrates them via subspace aggregation, facilitating inter-task knowledge sharing. However, linear subspace expansion leads to parameter redundancy as the task sequence grows. MOS [28] mitigates forgetting through "surgical" backbone modifications—such as inserting, pruning, and merging adapters—but relies on manually designed architectures, which hinders scalability. Beyond these, RanPAC [18] employs frozen random projections to enhance robustness, GC² [2] and DUCT [41] introduce expert subnetworks and dual consolidation mechanisms to preserve previously learned knowledge.

2.2. Parameter-Efficient Fine-Tuning (PEFT)

Parameter-Efficient Fine-Tuning (PEFT) improves PTM adaptability by introducing few trainable parameters while keeping the backbone frozen. Representative approaches include Adapters [24], Prompt-Tuning [13], and LoRA [11], which respectively modify intermediate layers, prepend learnable prompts, and adjust linear transformations. Among them, LoRA has gained wide adoption for its simplicity and scalability, decomposing weight updates into a product of two low-rank matrices to enable efficient fine-tuning with minimal overhead.

However, the performance of standard LoRA is highly sensitive to rank selection. Low ranks restrict expressiveness, whereas high ranks introduce redundancy and exacerbate catastrophic forgetting [3]. Recent advances have therefore focused on dynamic rank allocation. DoRA [17] decomposes LoRA layers into multiple rank-one components and prunes them adaptively based on importance. CoDyRA [15] employs rank-selective LoRA modules that adjust capacity according to data distributions, thereby balancing plasticity and stability. PEARL [3] further deter-

mines task-specific ranks by measuring parameter proximity to reference weights, mitigating forgetting while maintaining adaptability.

3. Method

3.1. Dynamic Domain-Incremental Learning

Traditional DIL typically assumes a fixed label space across domains, where the model adapts to distributional shifts while maintaining access to previously learned categories. However, this assumption rarely holds in real-world clinical scenarios. Different medical institutions may adopt distinct annotation protocols, diagnostic taxonomies, or inclusion criteria, leading to an evolving label space as new domains arrive. Consequently, the complete set of possible diagnostic categories is *unknown a priori*, making it infeasible to predefine a universal classifier encompassing all potential labels.

Formally, each domain t corresponds to a dataset $\mathcal{D}_t = \{(\mathbf{x}_i^{(t)}, y_i^{(t)})\}_{i=1}^{N_t}$ with its associated label set $\mathcal{Y}_t$. Unlike conventional DIL, the label space is allowed to evolve dynamically:

$$\mathcal{Y}_{t-1} \cap \mathcal{Y}_t \neq \emptyset, \quad \text{and} \quad |\mathcal{Y}_t \cup \mathcal{Y}_{t-1}| \geq |\mathcal{Y}_{t-1}|. \tag{1}$$

Eq. 1 reflects the practical setting where known categories persist while new ones emerge over time.

Rehearsal-free Constraint and Knowledge Retention. A key challenge in this dynamic DIL lies in the *rehearsal-free constraint*: once the model transitions to domain t, data from previous domains cannot be revisited due to privacy, storage, or ethical restrictions:

$$\mathcal{D}_i \cap \mathcal{D}_j = \emptyset, \quad \forall i \neq j. \tag{2}$$

Thus, the model is trained exclusively on $\mathcal{D}_t$ at each stage, while being required to retain competence across all previously encountered domains $\{\mathcal{D}_1, \ldots, \mathcal{D}_{t-1}\}$. This dual objective introduces a *stability–plasticity dilemma*, where the model must integrate new knowledge without overwriting previously learned discriminative representations.

Domain Formulation. Let $f_\theta(\cdot)$ denote the shared feature extractor parameterized by θ. At domain t, the classifier head is defined as

$$\mathbf{W}^{(t)} \in \mathbb{R}^{C_t \times d}, \quad C_t = \Big| \bigcup_{i=1}^{t} \mathcal{Y}_i \Big|, \tag{3}$$

where d denotes the feature dimension of the shared embedding space and C_t is the cumulative number of recognizable categories up to domain t. When new categories are introduced, the classifier expands accordingly:

$$\mathbf{W}^{(t)} = \begin{bmatrix} \mathbf{W}^{(t-1)} \\ \Delta\mathbf{W}^{(t)} \end{bmatrix}, \tag{4}$$

where $\Delta\mathbf{W}^{(t)}$ denotes the parameters associated with the newly added categories. The backbone network is kept frozen, and only the domain-adaptive modules are optimized on $\mathcal{D}_t$, so that the updated representation space remains compatible with the accumulated classifier weights $\{\mathbf{W}^{(i)}\}_{i<t}$. This dynamic formulation provides a unified abstraction of continual domain adaptation under evolving label spaces.

Prototype-based Classification. Unlike conventional linear classifiers that learn independent weight vectors, we interpret the classifier $\mathbf{W}^{(t)}$ as a *prototype memory bank* comprising the class prototypes accumulated up to domain t:

$$\mathbf{P}^{(t)} = \{\mathbf{p}_c \mid c \in \mathcal{Y}_1 \cup \cdots \cup \mathcal{Y}_t\}, \quad \mathbf{W}^{(t)} = [\mathbf{p}_1, \ldots, \mathbf{p}_{C_t}]. \tag{5}$$

Each prototype $\mathbf{p}_c$ corresponds to the normalized centroid of its class in the learned embedding space, i.e.,

$$\mathbf{p}_c^{(t)} = \frac{1}{|\bigcup_{k=1}^{t} \mathcal{D}_k^c|} \sum_{(\mathbf{x}_i, y_i) \in \bigcup_{k=1}^{t} \mathcal{D}_k^c} \frac{f_\theta(\mathbf{x}_i)}{\|f_\theta(\mathbf{x}_i)\|_2}, \tag{6}$$

which ensures accumulated prototype updates across domains. During inference, the model predicts the label of an input $\mathbf{x}$ based on cosine similarity to all stored prototypes:

$$\hat{y} = \arg\max_{c \in \mathcal{Y}_{1:t}} \cos\big(f_\theta(\mathbf{x}),\, \mathbf{p}_c\big), \tag{7}$$

where $\mathcal{Y}_{1:t}$ denotes the cumulative label set up to domain t. This prototype-based formulation naturally accommodates dynamic label expansion and facilitates contrastive refinement during subsequent KIR training.

3.2. Dynamic Adaptation Module (DAM)

LoRA has shown strong potential for incremental learning by introducing domain-specific low-rank updates to pre-trained weights while keeping the base model frozen [9].

However, conventional LoRA assumes a fixed latent rank throughout training, which is suboptimal in dynamic DIL scenarios, where domains differ in data scale, distribution, and label complexity. To address this, we propose the DAM, a rank-adaptive low-rank adapter that dynamically adjusts its internal rank and modulation strength, enabling domain-aware capacity scaling.

Integration with Transformer Blocks. We integrate DAM into the ViT backbone by attaching low-rank adapters to the existing linear projections within both self-attention and linear projection (*Proj.*) layers. Let $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ denote the original weight matrix of a linear projection, where d_{in} and d_{out} are the input and output feature dimensions, respectively. For each projection, DAM introduces a learnable residual branch:

$$W' = W + \Delta W, \quad \Delta W = AB, \tag{8}$$

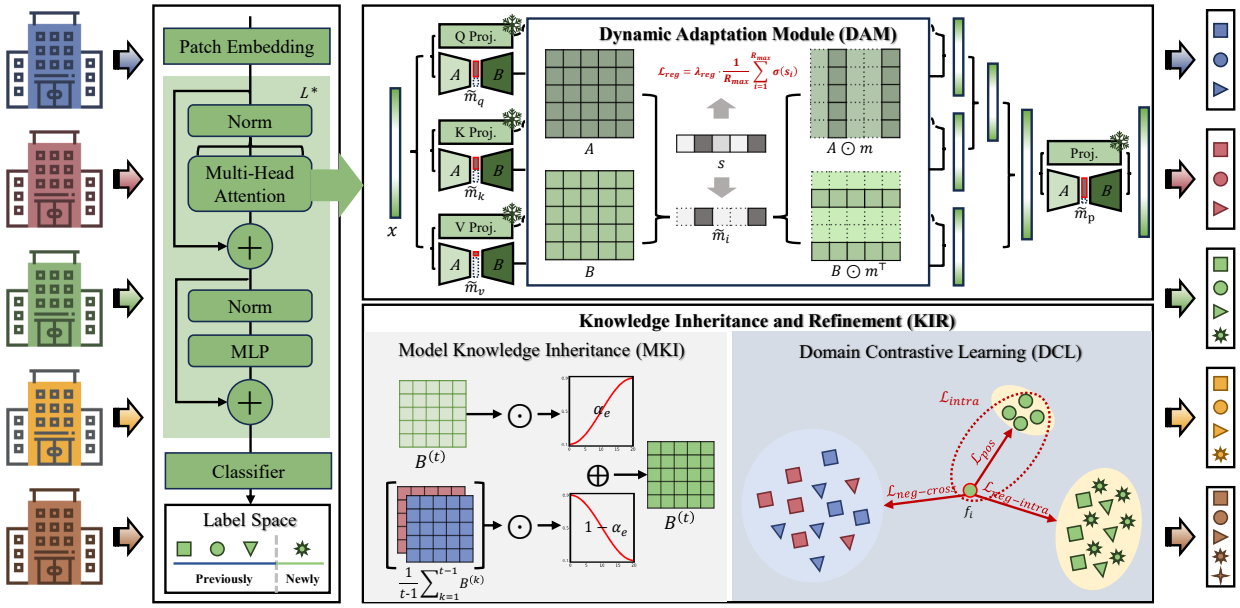

Figure 1. Overview of the proposed DK-DDIL framework. Given sequential domains with evolving label spaces, DK-DDIL enables rehearsal-free continual adaptation through two key components. (1) The **Dynamic Adaptation Module (DAM)** adaptively adjusts the low-rank capacity of LoRA-based adapters via learnable rank regulation, allowing domain-aware capacity scaling without modifying pretrained weights. (2) The **Knowledge Inheritance and Refinement (KIR)** mechanism preserves cross-domain consistency through model fusion based knowledge inheritance (MKI) and domain contrastive learning (DCL), which jointly mitigate catastrophic forgetting and enhance inter-domain discriminability.

where $A \in \mathbb{R}^{d_{\text{out}} \times r_{\max}}$ and $B \in \mathbb{R}^{r_{\max} \times d_{\text{in}}}$ represent the low-rank matrices introduced for adaptation, with $r_{\max}$ denoting the maximum latent rank shared across domains. This design allows each block to adaptively adjust the contribution of its query-key-value relations and *Proj.* transformations to new domains, without modifying the pretrained backbone weights. The integration ensures that the model can efficiently incorporate new domain-specific knowledge while preserving the pretrained representations.

Dynamic Rank Regulation. Each DAM maintains a learnable rank score vector $\mathbf{s} \in \mathbb{R}^{r_{\max}}$, whose elements determine the activation of latent dimensions via a temperature-scaled sigmoid function:

$$\tilde{m}_i = \sigma(s_i), \quad m_i = \mathbb{I}[\tilde{m}_i > \tau] + (\tilde{m}_i - \text{stopgrad}(\tilde{m}_i)). \tag{9}$$

where $\sigma(\cdot)$ denotes the sigmoid function, $\mathbb{I}[\cdot]$ is the indicator operator, and $\text{stopgrad}(\cdot)$ represents the stop-gradient operation used in the straight-through estimator (STE). Here, τ is a learnable threshold, and the STE enables discrete sampling $m_i \in \{0, 1\}$ while preserving differentiability during backpropagation. This process allows the model to dynamically select the most relevant rank components for each domain while preserving gradient flow for stable training.

To avoid excessive rank reduction and ensure baseline capacity, DAM enforces a minimum-rank constraint $r_{\min}$ as a deterministic fallback. Specifically, we ensure that the total number of activated dimensions is always above the minimum rank:

$$\sum_{i=1}^{r_{\max}} m_i \geq r_{\min}. \tag{10}$$

If the number of activated dimensions according to m falls below $r_{\min}$, the top-$r_{\min}$ score indices are forcibly activated. This guarantees that DAM always retains a compact but sufficient subspace for stable inference and optimization. The minimum rank $r_{\min}$ serves as a baseline to avoid excessive pruning of dimensions, ensuring that the model retains enough capacity to represent domain-specific features. Given the binary mask $\mathbf{m}$, the masked low-rank update is defined as:

$$\Delta W_m = A \, \text{diag}(\mathbf{m}) \, B, \tag{11}$$

where $\text{diag}(\mathbf{m})$ selects active latent components. To maintain balanced contribution between pretrained and adapted parameters, we apply a dynamic scaling factor α_t proportional to the effective rank:

$$\alpha_t = \frac{r_{\max}}{\sum_i m_i}, \tag{12}$$

$$h = Wx + \alpha_t \cdot \Delta W_m x. \tag{13}$$

where $x \in \mathbb{R}^{d_{\text{in}}}$ and $h \in \mathbb{R}^{d_{\text{out}}}$ denote the input and output features of a linear projection, respectively. This scaling ensures the adaptation strength aligns with the number of activated components, allowing domain-specific modulation of model capacity.

To prevent overfitting and encourage sparsity, we introduce a regularization loss on the rank scores:

$$\mathcal{L}_{\text{reg}} = \lambda_{\text{reg}} \cdot \frac{1}{r_{\max}} \sum_{i=1}^{r_{\max}} \sigma(s_i), \tag{14}$$

where λ_{reg} controls sparsity strength. This regularization encourages sparsity, prompting the model to activate only the most essential rank components for each domain.

Unlike existing LoRA or domain-wise rank tuning methods [9], which allocate ranks at a coarse domain level under fixed label spaces, our regulation mechanism learns continuous rank masks in a fully differentiable and data-adaptive manner. Specifically, the proposed DAM introduces per-domain, fine-grained rank modulation that enables smooth capacity adjustment guided by data statistics, rather than discrete rank switching. This design enables smooth and continuous capacity scaling across evolving domains, making DAM a principled, rank-adaptive solution for dynamic DIL.

3.3. Knowledge Inheritance and Refinement (KIR)

While DAM provides adaptive parameterization, it does not explicitly preserve inter-domain consistency. To ensure maintaining cross-domain consistency and prototype-level discriminability, we introduce the KIR mechanism, which comprises two cooperative components: (i) *Model Fusion based Knowledge Inheritance (MKI)* for parameter-level transfer, and (ii) *Domain Contrastive Learning (DCL)* for prototype-level refinement.

3.3.1. Model Fusion based Knowledge Inheritance (MKI)

Reusing past adapters in dynamic DIL often leads to interference due to domain misalignment. Inspired by model fusion techniques such as [19], we propose MKI, which dynamically merges the current DAM with historical DAMs to achieve smooth knowledge transfer and balance stability with plasticity.

The goal of MKI is twofold: to inherit domain-invariant priors from previous DAMs that capture stable and transferable structures, and to maintain inter-class separability by avoiding over-smoothing during parameter fusion.

Adaptive Fusion Scheduling. To balance knowledge retention and domain adaptation, we introduce a cosine-annealed coefficient α_e to control the strength of parameter inheritance across training epochs:

$$\{\alpha_{\text{init}}, \alpha_{\text{final}}\} \in [0,1]^2, \quad \alpha_{\text{final}} = 1 - \alpha_{\text{init}}, \tag{15}$$

$$\alpha_e = \alpha_{\text{final}} + (\alpha_{\text{init}} - \alpha_{\text{final}}) \cdot \frac{1 + \cos(\pi e / E)}{2}. \tag{16}$$

where α_{init} is a hyperparameter, e denotes the current epoch and E is the total number of epochs. This scheduling allows the model to emphasize knowledge inheritance at the beginning of training (large α_e) and progressively shift toward domain-specific learning as α_e decays. Compared with linear or exponential decay, the cosine-annealed schedule provides a smoother transition between stability and plasticity, reducing abrupt parameter changes and improving convergence stability—an especially desirable property under the rehearsal-free constraint of dynamic DIL.

Selective Parameter Fusion. While the adaptive fusion scheduling determines *when* and *how strongly* inheritance occurs, it does not specify *which components* of the DAM should participate in fusion. To stabilize cross-domain transfer, we restrict model inheritance to low-rank components that encode relatively domain-invariant relations. Let $\{\mathcal{A}^{(1)}, \ldots, \mathcal{A}^{(t-1)}\}$ denote the previously learned DAMs and $\mathcal{A}^{(t)}$ the current one. We perform parameter-level fusion only on the projection matrices B, which capture the global subspace structure of feature interactions and have been empirically observed to generalize more consistently across domains:

$$B^{(t)} \leftarrow \alpha_e \, B^{(t)} + \frac{(1-\alpha_e)}{t-1} \sum_{k=1}^{t-1} B^{(k)}, \tag{17}$$

where $B^{(t)}$ denotes the low-rank projection in the current DAM and α_e is the adaptive fusion coefficient at epoch e.

This selective fusion focuses on the *inherently stable subspace* of DAM while keeping the complementary components independently optimized for domain-specific plasticity. In this way, MKI achieves a principled balance between knowledge inheritance and domain adaptation without disrupting domain-specialized representations.

3.3.2. Knowledge Refinement via Domain Contrastive Learning (DCL)

Although the MKI mechanism effectively stabilizes parameter adaptation across domains, feature embeddings may still experience *prototype drift* (shifts of class centroids) in embedding space and *semantic confusion* (misalignment of features across domains). This occurs because newly adapted representations can inadvertently align with outdated prototypes from previous domains, especially when label spaces partially overlap or feature subspaces evolve. Such misalignment degrades inter-domain consistency and reduces discriminative power, which is particularly detrimental in dynamic DIL, where replay data are unavailable for correction. To further refine discriminative structure, we design a DCL objective that jointly enforces intra-class compactness, inter-class separation, and cross-domain confusion suppression.

Positive Alignment. Given a mini-batch of normalized features $\mathbf{F} = \{\mathbf{f}_i\}_{i=1}^{\mathbb{B}} \in \mathbb{R}^{\mathbb{B} \times d}$ and the current-domain proto-

types $\mathbf{P}^{(t)} = \{\mathbf{p}_k\}_{k=1}^{C_t} \in \mathbb{R}^{C_t \times d}$, DCL first encourages each feature to align with its corresponding class prototype:

$$\mathcal{L}_{\text{pos}} = \frac{1}{\mathbb{B}} \sum_{i=1}^{\mathbb{B}} \left[1 - \cos(\mathbf{f}_i, \mathbf{p}_{y_i}^{(t)})\right], \tag{18}$$

where $\mathbb{B}$ denotes the mini-batch size, $\mathbf{f}_i \in \mathbb{R}^d$ is the feature of the i-th sample, and $y_i \in C_t$ is its corresponding class label. Both $\mathbf{f}_i$ and $\mathbf{p}_{y_i}^{(t)}$ are ℓ_2-normalized. This term enhances intra-class consistency and mitigates the noise introduced by prototype drift or unstable class centroids.

Intra-domain Contrastive Separation. For inter-class separation within the current domain, DCL introduces an InfoNCE-style contrastive loss over prototype similarities:

$$\mathcal{L}_{\text{neg-intra}} = -\frac{1}{\mathbb{B}} \sum_{i=1}^{\mathbb{B}} \log \frac{\exp(\cos(\mathbf{f}_i, \mathbf{p}_{y_i}^{(t)}))}{\sum_{j=1}^{C_t} \exp(\cos(\mathbf{f}_i, \mathbf{p}_j^{(t)}))}. \tag{19}$$

Unlike conventional hard-negative mining, which can destabilize training under domain shifts, this formulation leverages prototype-level contrast, providing a more stable and domain-aware separation signal.

Cross-domain Negative Suppression. In dynamic DIL, samples from new domains may inadvertently align with *different-class prototypes* from earlier domains, leading to false positive associations during inference. To mitigate this, DCL explicitly penalizes such cross-domain confusions. Given previous-domain prototypes $\mathbf{P}^{(t-1)}$ and their associated labels $\mathcal{Y}_{t-1}$, we define:

$$g(\mathbf{f}, \mathbf{p}) = \exp\big(\cos(\mathbf{f}, \mathbf{p})\big), \tag{20}$$

$$\mathcal{L}_{\text{neg-cross}} = -\frac{1}{\mathbb{B}} \sum_{i=1}^{\mathbb{B}} \log \frac{g(\mathbf{f}_i, \mathbf{p}_{y_i}^{(t)})}{\sum_j g(\mathbf{f}_i, \mathbf{p}_j^{(t-1)}) \mathbb{1}\left[y_i \neq c_j^{(t-1)}\right]}, \tag{21}$$

where $c_j^{(t-1)}$ is the class label associated with $\mathbf{p}_j^{(t-1)}$ and $\mathbb{1}[\cdot]$ masks same-class prototypes to preserve semantic consistency. $\mathcal{L}_{\text{neg-cross}}$ encourages new-domain features to stay distinct from semantically unrelated historical prototypes, reducing inter-domain misclassification while maintaining class continuity across overlapping label spaces.

Intra-class Compactness. Beyond prototype alignment, to ensure local feature coherence independent of prototype stability, we impose an intra-class compactness constraint:

$$\mathcal{L}_{\text{intra}} = \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} [1 - \cos(\mathbf{f}_i, \mathbf{f}_j)], \tag{22}$$

where $\mathcal{P} = \{(i,j) \,|\, y_i = y_j,\, i \neq j\}$ represents pairs of samples within the same class in the current batch. This encourages feature cohesion at the instance level, complementing prototype-based alignment.

The overall DCL objective integrates multiple consistency terms under a curriculum-weighted formulation:

$$\mathcal{L}_{\text{DCL}} = \mathcal{L}_{\text{pos}} + \frac{s}{S_t}\big(\mathcal{L}_{\text{neg-intra}} + \mathcal{L}_{\text{neg-cross}}\big) + \mathcal{L}_{\text{intra}}. \tag{23}$$

Here, s denotes the current optimization step within domain t, and S_t represents the total number of samples (or iterations) associated with that domain. This formulation gradually strengthens the impact of negative contrastive regularization as more samples are observed.

Final Objective. The overall training objective is:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{reg}} + \mathcal{L}_{\text{DCL}}, \tag{24}$$

where $\mathcal{L}_{\text{CE}}$ denotes the cross-entropy loss. This joint objective enables dynamic capacity scaling and cross-domain representation alignment across evolving domains.

Different from prior adapter fusion or distillation-based DIL, KIR operates at two complementary levels. First, MKI selectively merges domain-invariant low-rank components guided by adaptive scheduling, preventing interference across evolving domains. Second, DCL refines prototype consistency at the representation level, mitigating prototype drift and inter-domain confusion. Together, MKI and DCL form a dual-consistency mechanism that stabilizes adaptation while enhancing cross-domain discriminability.

4. Experiments

4.1. Experimental Setup

We evaluate DK-DDIL across three representative scenarios to comprehensively assess its generalization and adaptability: (1) **Skin Pathology Diagnosis** [4, 10, 12, 20, 21, 29],

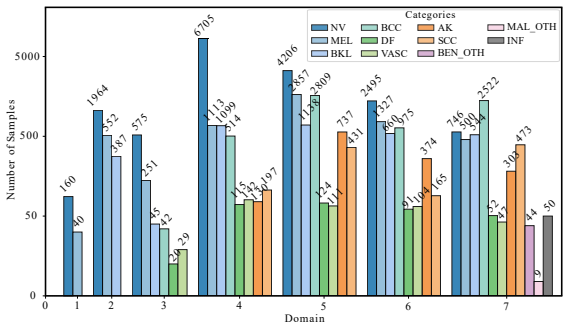

Figure 2. Sample distribution across seven sequential domains in the **Skin Pathology Diagnosis** setting. Each domain corresponds to a distinct clinical dataset or acquisition period, representing the temporal evolution of real-world dermatopathology practice. The y-axis is plotted on a logarithmic scale to emphasize cross-domain imbalance. Both the category composition and the sample size vary notably across domains—new lesion types (e.g., AK, SCC, MAL_OTH) appear in later stages, whereas earlier domains contain fewer samples with a more limited set of lesion categories.

Table 1. Quantitative comparison of state-of-the-art DIL methods on Skin Pathology Diagnosis, Cyst-X, and OfficeHome benchmarks. We report the average accuracy $\overline{A}$ and final accuracy A_T, along with the proportion of trainable parameters (%) and statistical significance (p-value).

Methods	Param (%)	p-value	Skin Pathology Diagnosis		Cyst-X		OfficeHome	
			$\overline{A}$	A_T	$\overline{A}$	A_T	$\overline{A}$	A_T
Finetune	100.00	< 0.05	$68.77_{\pm 5.39}$	$67.60_{\pm 4.34}$	$33.45_{\pm 7.97}$	$23.02_{\pm 13.32}$	$78.38_{\pm 0.50}$	$79.85_{\pm 0.49}$
L2P[33]	**0.15**	< 0.05	$69.94_{\pm 1.73}$	$64.54_{\pm 1.33}$	$31.60_{\pm 6.55}$	$32.37_{\pm 8.48}$	$78.80_{\pm 0.41}$	$81.24_{\pm 0.23}$
DualPrompt[32]	0.39	< 0.01	$72.58_{\pm 0.66}$	$67.06_{\pm 1.14}$	$49.32_{\pm 4.08}$	$49.64_{\pm 10.30}$	$77.30_{\pm 0.36}$	$80.42_{\pm 0.18}$
CODA-Prompt[26]	4.37	< 0.01	$73.17_{\pm 0.38}$	$67.11_{\pm 0.58}$	$49.02_{\pm 2.20}$	$48.92_{\pm 5.22}$	$81.37_{\pm 0.24}$	$84.18_{\pm 0.26}$
RanPAC[18]	2.03	< 0.01	$74.79_{\pm 0.10}$	$66.89_{\pm 0.09}$	$52.56_{\pm 1.39}$	$48.20_{\pm 4.14}$	$82.22_{\pm 0.09}$	$84.70_{\pm 0.12}$
EASE[38]	1.42	< 0.05	$54.02_{\pm 0.14}$	$41.11_{\pm 0.15}$	$38.11_{\pm 2.34}$	$41.01_{\pm 2.21}$	$80.96_{\pm 0.10}$	$83.37_{\pm 0.09}$
SimpleCIL[40]	100.00	< 0.05	$52.08_{\pm 0.00}$	$41.56_{\pm 0.00}$	$41.97_{\pm 1.30}$	$42.45_{\pm 1.17}$	$75.40_{\pm 0.00}$	$78.44_{\pm 0.00}$
Aper w/ VPT-Shallow[40]	41.68	< 0.01	$59.64_{\pm 3.29}$	$48.07_{\pm 2.36}$	$42.80_{\pm 4.70}$	$42.45_{\pm 3.40}$	$70.34_{\pm 2.80}$	$74.78_{\pm 2.23}$
Aper w/ VPT-Deep[40]	41.70	< 0.05	$56.08_{\pm 2.59}$	$41.11_{\pm 2.58}$	$42.77_{\pm 2.83}$	$41.73_{\pm 3.43}$	$78.65_{\pm 0.11}$	$80.89_{\pm 0.13}$
Aper w/ SSF[40]	41.90	< 0.05	$53.61_{\pm 0.62}$	$43.14_{\pm 0.50}$	$41.43_{\pm 1.65}$	$36.69_{\pm 1.55}$	$79.59_{\pm 0.08}$	$81.86_{\pm 0.06}$
Aper w/ Finetune[40]	100.00	< 0.01	$51.62_{\pm 1.31}$	$41.99_{\pm 1.49}$	$42.26_{\pm 0.55}$	$41.01_{\pm 1.26}$	$65.96_{\pm 0.50}$	$71.43_{\pm 0.43}$
Aper w/ Adapter[40]	1.81	< 0.01	$52.08_{\pm 0.02}$	$41.61_{\pm 0.03}$	$40.61_{\pm 1.70}$	$38.85_{\pm 1.66}$	$75.82_{\pm 0.03}$	$78.82_{\pm 0.02}$
DGR[8]	100.00	< 0.01	$64.59_{\pm 1.71}$	$45.88_{\pm 2.32}$	$34.14_{\pm 8.78}$	$26.62_{\pm 9.53}$	$74.55_{\pm 0.28}$	$66.00_{\pm 0.49}$
GC2[2]	100.00	< 0.01	$72.11_{\pm 4.45}$	$68.70_{\pm 6.00}$	$44.32_{\pm 8.54}$	$45.47_{\pm 11.22}$	$77.88_{\pm 0.32}$	$80.89_{\pm 0.73}$
MOS[28]	1.35	< 0.05	$71.48_{\pm 0.36}$	$63.67_{\pm 0.12}$	$38.63_{\pm 1.37}$	$31.65_{\pm 3.88}$	$79.46_{\pm 0.40}$	$83.12_{\pm 0.30}$
DUCT[41]	100.00	< 0.05	$71.44_{\pm 0.89}$	$66.40_{\pm 1.29}$	$52.51_{\pm 2.19}$	$49.64_{\pm 1.41}$	$81.88_{\pm 0.11}$	$85.80_{\pm 0.21}$
CL-LoRA[9]	0.62	< 0.05	$72.53_{\pm 0.86}$	$68.91_{\pm 0.47}$	$40.82_{\pm 1.86}$	$33.09_{\pm 3.85}$	$79.20_{\pm 0.22}$	$84.04_{\pm 0.22}$
DK-DDIL (Ours)	**0.26**	-	$\mathbf{77.03_{\pm 0.52}}$	$\mathbf{71.52_{\pm 1.04}}$	$\mathbf{53.34_{\pm 3.98}}$	$\mathbf{51.08_{\pm 4.74}}$	$\mathbf{84.35_{\pm 0.30}}$	$\mathbf{86.29_{\pm 0.19}}$

constructed from multiple public dermoscopic datasets reflecting the temporal evolution of clinical practice, with the sample distribution illustrated in Fig. 2; (2) **Cyst-X** [23], a representative multi-center MRI dataset covering varied acquisition protocols and patient populations; and (3) **OfficeHome** [30], a standard non-medical benchmark to test generalization beyond clinical domains. These settings jointly cover dynamic clinical and standard benchmarks, providing a comprehensive evaluation under both realistic and controlled conditions.

Comparison Methods. We include both prompt-based and adapter-based DIL approaches, such as L2P [33], DualPrompt [32], CODA-Prompt [26], EASE [38], RanPAC [18], MOS [28], CL-LoRA [9], DUCT [41], SimpleCIL [40], and APER [40], which adopt lightweight integration strategies such as VPT, SSF, Finetune, and Adapter variants. Additionally, exemplar-based methods such as DGR [8] and pruning-based approaches like GC2 [2] are included to evaluate generalization to domain-specific continual learning settings, thereby enabling a direct comparison between DK-DDIL and these approaches tailored for clinical scenarios.

Implementation Details. Following DUCT [41] and CL-LoRA [9], we adopt ViT-B/16 [6] with $N = 12$ transformer blocks pretrained on ImageNet-21K [5] as the backbone architecture for all experiments. Specifically, the rank is dynamically adjusted between $r_{\min} = 4$ and $r_{\max} = 128$ according to the proposed DAM mechanism. The regularization coefficient is set to $\lambda_{\text{reg}} = 1$. The adaptive inheritance factor is initialized as $\alpha_{\text{init}} = 0.1$ and updated following Eq. 16. All methods are implemented within the LAMDA-PILOT framework [27, 37, 39], except GC2 [2] and CL-LoRA [9], which are reproduced based on their official implementations. This ensures fair and consistent evaluation. Results are averaged over five runs.

4.2. Benchmark Comparison

As shown in Table 1, DK-DDIL consistently achieves superior performance across all benchmarks, demonstrating its robustness under diverse DIL scenarios. On **Skin Pathology Diagnosis**, DK-DDIL achieves the best average and final accuracies, outperforming both prompt- and adapter-based baselines. The gain primarily stems from the proposed DAM and KIR modules, which better capture temporal domain shifts inherent to clinical data. For the **Cyst-X** dataset, which poses significant cross-institutional domain gaps, DK-DDIL maintains strong generalization and surpasses the second-best method by nearly 1% in both $\overline{A}$ and A_T. This demonstrates that DK-DDIL effectively mitigates feature drift across evolving domains without catastrophic forgetting. On the **OfficeHome** benchmark, DK-DDIL achieves comparable performance to existing methods with remarkable parameter efficiency ($< 0.3\%$ trainable parameters), nearly one order of magnitude fewer than typical baselines. This highlights that DK-DDIL not only preserves general knowledge from pre-trained models but also provides efficient domain-specific adaptation.

Notably, paired t-tests ($p < 0.05$) confirm the statistical significance and consistency of the observed improvements across repeated runs. Overall, these results validate that DK-DDIL achieves a favorable balance between accuracy, adaptability, and parameter efficiency, effectively addressing the challenges of DIL across both medical and natural image settings.

4.3. Ablation Study

We conduct extensive ablation studies to analyze the contribution of each component in our DK-DDIL framework, as

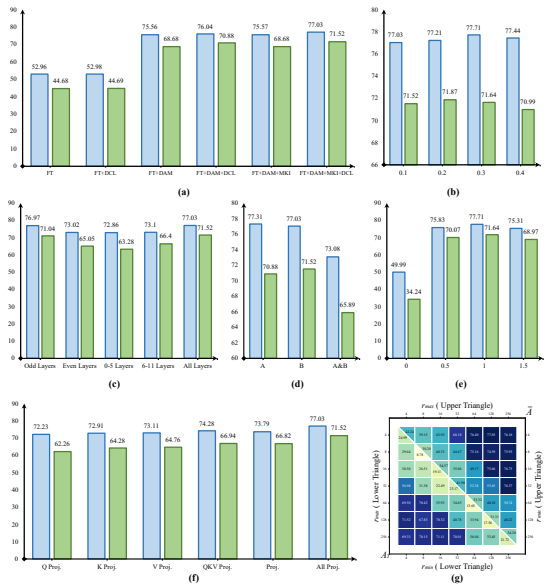

Figure 3. **Ablation study of DK-DDIL.** (a) Overall contribution on the key components. "FT" denotes fine-tuning only the classifier with frozen backbone. (b) Influence of α_{init} on the DK-DDIL's stability. (c) Impact of inserting DAM into different ViT layers (Odd / Even / 0-5 / 6-11 / All). (d) Effect of different parameter settings within MKI. (e) Sensitivity of DK-DDIL to the regularization weight λ_{reg}. (f) Comparison of DAM injected into different positions of the self-attention mechanism (Q, K, V, *Proj.*, and All). (g) Effect of different $(r_{\min}, r_{\max})$ configurations. In (a)–(f), blue and green bars represent $\overline{A}$ and A_T, respectively. In (g), the upper and lower triangles correspond to $\overline{A}$ and A_T.

illustrated in Fig. 3.

As shown in Fig. 3 (a), FT suffers from significant performance degradation due to the absence of DAM, MKI, and DCL. Introducing DCL slightly improves performance but remains limited without domain-aware adaptation. By incorporating DAM, the model effectively learns domain adaptation representations, achieving a notable performance boost. MKI further enhances performance by promoting mutual knowledge transfer between domain-adapted and shared representations (i.e., aligning new domain features with the general knowledge inherited from previous domains). When all components are combined, DK-DDIL achieves the best performance, confirming the complementary nature of these components.

Figure 3 (b) shows that $\alpha_{\text{init}} = 0.3$ yields the best performance, while too small or large values lead to under- or over-adaptation. As illustrated in Fig. 3 (c), inserting DAM into all layers achieves the best overall performance by harmonizing both low-level and high-level feature adaptation. Interestingly, we observe that applying DAM **only** to the odd layers can also yield a comparable result, indicating that evenly distributed but sparser adaptation is sufficient to maintain domain adaptation while reducing computational cost. This suggests that alternating-layer insertion offers a good trade-off between efficiency and representation capacity. In Fig. 3 (d), we analyze different parameter configurations within MKI. Combining multiple parameter sets (*A&B*) does not further improve performance, indicating that a single, well-chosen configuration is sufficient for stable integration. As shown in Fig. 3 (e), moderate values of λ_{reg} yield the best results. An excessively small value weakens the regularization effect, while too large a value constrains model flexibility. Figure 3 (f) compares different positions for DAM insertion. Injecting DAM across all projection layers ("All Proj.") provides the most consistent improvement, suggesting that domain adaptation benefits from multi-branch interaction.

Finally, Fig. 3 (g) analyzes the influence of the rank range $(r_{\min}, r_{\max})$ on model performance. We explore various combinations and observe that DK-DDIL maintains stable accuracy across a wide range of configurations, particularly when $r_{\min}$ remains relatively small, suggesting robustness to moderate variations in the lower-rank bound. As $r_{\min}$ increases further, the benefit of larger ranks gradually diminishes, likely due to reduced regularization capacity and weaker suppression of redundant domain-specific variations. Meanwhile, increasing $r_{\max}$ generally improves the representational capacity of DAM, but overly large values may introduce redundant subspaces and weaken generalization. Overall, a moderate rank range (e.g., small $r_{\min}$ and reasonably large $r_{\max}$) provides a good balance between adaptation flexibility and stability, demonstrating that DAM performs robustly across diverse domain shifts.

5. Conclusion and Future works

In this paper, we propose **DK-DDIL**, a rehearsal-free Dynamic DIL framework designed for efficient and flexible adaptation across evolving domains. DK-DDIL integrates DAM and KIR as two complementary components, jointly enabling robust and dynamic knowledge adaptation. This synergy empowers DK-DDIL to maintain both plasticity and stability, offering a general paradigm for efficient adaptation under dynamic and non-stationary domain shifts. Comprehensive experiments on both medical and natural image benchmarks demonstrate the robustness and versatility of DK-DDIL. These results highlight DK-DDIL as a promising step toward scalable and privacy-preserving DIL systems, particularly suited for real-world clinical deployment where data distributions evolve over time and across institutions.

In future work, we plan to extend DK-DDIL to more complex scenarios such as cross-modal continual learning. We also aim to explore integrating DK-DDIL with foundation models for advancing toward lifelong and trustworthy AI in dynamic medical environments.

Acknowledgments

This work was supported by the Ministry of Science and Technology of the People's Republic of China (STI2030-Major Projects 2021ZD0201900) and the National Natural Science Foundation of China (Grant No. 62371409).

References

[1] Bobby Azad, Reza Azad, Sania Eskandari, Afshin Bozorgpour, Amirhossein Kazerouni, Islem Rekik, and Dorit Merhof. Foundational models in medical imaging: A comprehensive survey and future vision, 2023. 1

[2] Nourhan Bayasi, Ghassan Hamarneh, and Rafeef Garbi. Gc2: Generalizable continual classification of medical images. *IEEE Transactions on Medical Imaging*, 43(11):3767–3779, 2024. 2, 7

[3] Prashant Shivaram Bhat, Shakib Yazdani, Elahe Arani, and Bahram Zonooz. Parameter efficient continual learning with dynamic low-rank adaptation, 2025. 2

[4] Noel C. F. Codella, David Gutman, M. Emre Celebi, Brian Helba, Michael A. Marchetti, Stephen W. Dusza, Aadi Kalloo, Konstantinos Liopyris, Nabin Mishra, Harald Kittler, and Allan Halpern. Skin lesion analysis toward melanoma detection: A challenge at the 2017 international symposium on biomedical imaging (isbi), hosted by the international skin imaging collaboration (isic), 2018. 6

[5] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*, pages 248–255. Ieee, 2009. 7

[6] Alexey Dosovitskiy. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020. 7

[7] Hao Guan and Mingxia Liu. Domain adaptation for medical image analysis: a survey. *IEEE Transactions on Biomedical Engineering*, 69(3):1173–1185, 2021. 1

[8] Jiangpeng He. Gradient reweighting: Towards imbalanced class-incremental learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16668–16677, 2024. 7

[9] Jiangpeng He, Zhihao Duan, and Fengqing Zhu. Cl-lora: Continual low-rank adaptation for rehearsal-free class-incremental learning. *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR)*, pages 30534–30544, 2025. 3, 5, 7

[10] Cristina Hernández-Pérez, Marc Combalia, Sebastian Podlipnik, Verónica Vilaplana, Susana Puig, and Josep Malvehy. BCN20000: Dermoscopic Lesions in the Wild. *Scientific Data*, 11:641, 2024. 6

[11] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. *ICLR*, 1(2):3, 2022. 2

[12] Jeremy Kawahara, Sara Daneshvar, Giuseppe Argenziano, and Ghassan Hamarneh. Seven-point checklist and skin lesion classification using multitask multimodal neural nets. *IEEE Journal of Biomedical and Health Informatics*, 23(2): 538–546, 2019. 6

[13] Brian Lester, Rami Al-Rfou, and Noah Constant. The power of scale for parameter-efficient prompt tuning. *arXiv preprint arXiv:2104.08691*, 2021. 2

[14] Chenxin Li, Xin Lin, Yijin Mao, Wei Lin, Qi Qi, Xinghao Ding, Yue Huang, Dong Liang, and Yizhou Yu. Domain generalization on medical imaging classification using episodic training with task augmentation. *Computers in biology and medicine*, 141:105144, 2022. 1

[15] Haodong Lu, Chongyang Zhao, Jason Xue, Lina Yao, Kristen Moore, and Dong Gong. Adaptive rank, reduced forgetting: Knowledge retention in continual learning vision-language models with dynamic rank-selective lora. *arXiv preprint arXiv:2412.01004*, 2024. 2

[16] Yuxi Ma, Jiacheng Wang, Jing Yang, and Liansheng Wang. Model-heterogeneous semi-supervised federated learning for medical image segmentation. *IEEE Transactions on Medical Imaging*, 43(5):1804–1815, 2024. 1

[17] Yulong Mao, Kaiyu Huang, Changhao Guan, Ganglin Bao, Fengran Mo, and Jinan Xu. Dora: Enhancing parameter-efficient fine-tuning with dynamic rank distribution, 2024. 2

[18] Mark D McDonnell, Dong Gong, Amin Parvaneh, Ehsan Abbasnejad, and Anton Van den Hengel. Ranpac: Random projections and pre-trained models for continual learning. *Advances in Neural Information Processing Systems*, 36:12022–12053, 2023. 1, 2, 7

[19] Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Aguera y Arcas. Communication-efficient learning of deep networks from decentralized data. In *Artificial intelligence and statistics*, pages 1273–1282, 2017. 5

[20] Teresa Mendonça, Pedro M. Ferreira, Jorge Marques, André R. S. Marçal, and Jorge Rozeira. Ph² - a dermoscopic image database for research and benchmarking. In *Proceedings of the 35th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pages 5437–5440, Osaka, Japan, 2013. IEEE. 6

[21] MILK study team. MILK10k, 2025. ISIC Archive, Data set. 6

[22] Michael Moor, Oishi Banerjee, Zahra Shakeri Hossein Abad, Harlan M Krumholz, Jure Leskovec, Eric J Topol, and Pranav Rajpurkar. Foundation models for generalist medical artificial intelligence. *Nature*, 616(7956):259–265, 2023. 1

[23] Hongyi Pan, Gorkem Durak, Elif Keles, Deniz Seyithanoglu, Zheyuan Zhang, Alpay Medetalibeyoglu, Halil Ertugrul Aktas, Andrea Mia Bejar, Ziliang Hong, Yavuz Taktak, Gulbiz Dagoglu Kartal, Mehmet Sukru Erturk, Timurhan Cebeci, Maria Jaramillo Gonzalez, Yury Velichko, Lili Zhao, Emil Agarunov, Federica Proietto Salanitri, Concetto Spampinato, Pallavi Tiwari, Ziyue Xu, Sachin Jambawalikar, Ivo G. Schoots, Marco J. Bruno, Chenchang Huang, Candice W. Bolan, Tamas Gonda, Frank H. Miller, Rajesh N. Keswani, Michael B. Wallace, and Ulas Bagci. Cyst-x: A federated ai system outperforms clinical guidelines to detect pancreatic cancer precursors and reduce unnecessary surgery, 2025. 7

[24] Clifton Poth, Hannah Sterz, Indraneil Paul, Sukannya Purkayastha, Leon Engländer, Timo Imhof, Ivan Vulić, Sebastian Ruder, Iryna Gurevych, and Jonas Pfeiffer. Adapters: A unified library for parameter-efficient and modular transfer learning. *arXiv preprint arXiv:2311.11077*, 2023. 2

[25] Mohammad Areeb Qazi, Anees Ur Rehman Hashmi, Santosh Sanjeev, Ibrahim Almakky, Numan Saeed, Camila Gonzalez, and Mohammad Yaqub. Continual learning in medical imaging: A survey and practical analysis. *arXiv preprint arXiv:2405.13482*, 2024. 1

[26] James Seale Smith, Leonid Karlinsky, Vyshnavi Gutta, Paola Cascante-Bonilla, Donghyun Kim, Assaf Arbelle, Rameswar Panda, Rogerio Feris, and Zsolt Kira. Coda-prompt: Continual decomposed attention-based prompting for rehearsal-free continual learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 11909–11919, 2023. 1, 2, 7

[27] Hai-Long Sun, Da-Wei Zhou, De-Chuan Zhan, and Han-Jia Ye. Pilot: A pre-trained model-based continual learning toolbox. *SCIENCE CHINA Information Sciences*, 68(4):147101, 2025. 7

[28] Hai-Long Sun, Da-Wei Zhou, Hanbin Zhao, Le Gan, De-Chuan Zhan, and Han-Jia Ye. Mos: Model surgery for pre-trained model-based class-incremental learning. In *AAAI*, pages 20699–20707, 2025. 2, 7

[29] Philipp Tschandl, Cliff Rosendahl, and Harald Kittler. The ham10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions. *Scientific Data*, 5:180161, 2018. 6

[30] Hemanth Venkateswara, Jose Eusebio, Shayok Chakraborty, and Sethuraman Panchanathan. Deep hashing network for unsupervised domain adaptation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 5018–5027, 2017. 7

[31] Yiming Wang, Yu Lin, Xiaodong Zeng, and Guannan Zhang. Multilora: Democratizing lora for better multi-task learning, 2023. 2

[32] Zifeng Wang, Zizhao Zhang, Sayna Ebrahimi, Ruoxi Sun, Han Zhang, Chen-Yu Lee, Xiaoqi Ren, Guolong Su, Vincent Perot, Jennifer Dy, et al. Dualprompt: Complementary prompting for rehearsal-free continual learning. In *European conference on computer vision*, pages 631–648. Springer, 2022. 1, 2, 7

[33] Zifeng Wang, Zizhao Zhang, Chen-Yu Lee, Han Zhang, Ruoxi Sun, Xiaoqi Ren, Guolong Su, Vincent Perot, Jennifer Dy, and Tomas Pfister. Learning to prompt for continual learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 139–149, 2022. 1, 2, 7

[34] Xinyao Wu, Zhe Xu, and Raymond Kai-yu Tong. Continual learning in medical image analysis: A survey. *Computers in Biology and Medicine*, 182:109206, 2024. 1

[35] Yaming Yang, Dilxat Muhtar, Yelong Shen, Yuefeng Zhan, Jianfeng Liu, Yujing Wang, Hao Sun, Denvy Deng, Feng Sun, Qi Zhang, Weizhu Chen, and Yunhai Tong. Mtl-lora: Low-rank adaptation for multi-task learning, 2025. 2

[36] Kai Zhang, Rong Zhou, Eashan Adhikarla, Zhiling Yan, Yixin Liu, Jun Yu, Zhengliang Liu, Xun Chen, Brian D Davison, Hui Ren, et al. A generalist vision–language foundation model for diverse biomedical tasks. *Nature Medicine*, 30 (11):3129–3141, 2024. 1

[37] Da-Wei Zhou, Hai-Long Sun, Jingyi Ning, Han-Jia Ye, and De-Chuan Zhan. Continual learning with pre-trained models: A survey. In *IJCAI*, pages 8363–8371, 2024. 7

[38] Da-Wei Zhou, Hai-Long Sun, Han-Jia Ye, and De-Chuan Zhan. Expandable subspace ensemble for pre-trained model-based class-incremental learning. In *CVPR*, pages 23554–23564, 2024. 2, 7

[39] Da-Wei Zhou, Qi-Wei Wang, Zhi-Hong Qi, Han-Jia Ye, De-Chuan Zhan, and Ziwei Liu. Class-incremental learning: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(12):9851–9873, 2024. 7

[40] Da-Wei Zhou, Zi-Wen Cai, Han-Jia Ye, De-Chuan Zhan, and Ziwei Liu. Revisiting class-incremental learning with pre-trained models: Generalizability and adaptivity are all you need. *International Journal of Computer Vision*, 133(3): 1012–1032, 2025. 1, 7

[41] Da-Wei Zhou, Zi-Wen Cai, Han-Jia Ye, Lijun Zhang, and De-Chuan Zhan. Dual consolidation for pre-trained model-based domain-incremental learning. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 20547–20557, 2025. 1, 2, 7